

Part 1: Basic Discrete Probability

Example: A European call option with strike price K and maturity T gives the holder the right to purchase a share of said stock for K dollars at time T .

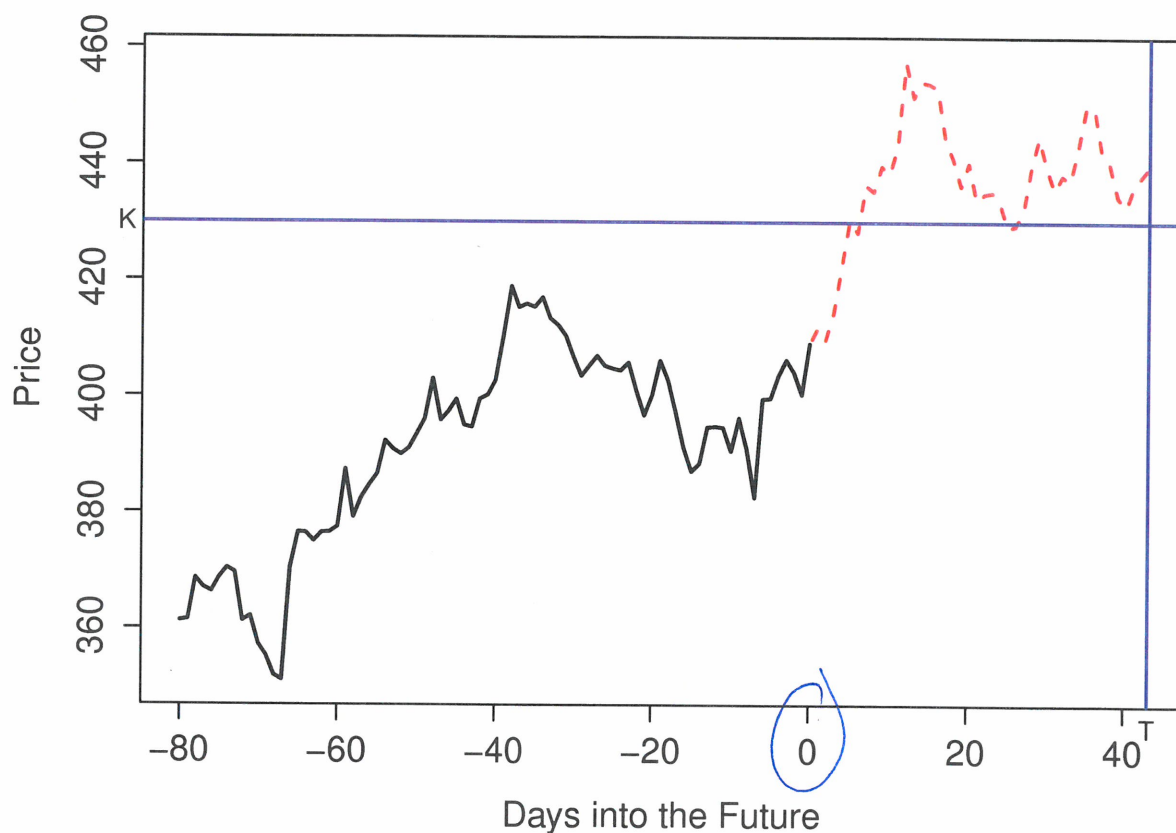


Figure 1: An example where $K = 430$ and $T = 43$. Time is adjusted so that the present is time zero. (This is GOOG from January through June, 2013.)

What is a reasonable premium to pay for this option?

$$X_T = \text{price of stock at time } T$$

If $X_T > k$, can exercise and make $X_T - k$

If $X_T \leq k$, option is worthless

So, not willing to pay more than

$$(X_T - k)^+$$

$$\text{where } Y^+ = \begin{cases} Y, & Y \geq 0 \\ 0, & \text{otherwise} \end{cases}$$

In fact, should not pay more than

$$e^{-rT} (X_T - k)^+$$

where r = risk free rate of return

Of course, I do not know X_T . Two alternatives:

① Try to predict X_T . Think about market, etc.

② Put forth a probability model for X_T .
(A process whose value is determined by chance.) that captures the variability and drift one would expect in a price process

Then calculate

$$E \left[e^{-rT} (X_T - k)^+ \right]$$

Approach ② is what we are interested in.

Comments:

① The prob. model also gives me

$$V(e^{-rT} (X_T - k)^+)$$

and the distribution of $e^{-rT} (X_T - k)^+$,
and calculate

$$P(e^{-rT} (X_T - k)^+ \leq c) \quad \text{for some } c$$

$$\textcircled{2} \quad E\left[e^{-rT}(X_T - k)^+\right] \neq e^{-rT}(E(X_T) - k)^+$$

In fact,

$$E\left[e^{-rT}(X_T - k)^+\right] \geq e^{-rT}(E(X_T) - k)^+$$

by Jensen's inequality

$\textcircled{3}$ As stated above, this model should capture variability and drift in real price. How do we do this in practice?

In practice, prob. models have unknown parameters which are estimated using historical data (FDS I and II)
"model calibration"

We should always be aware of the assumptions being made by model
Are they realistic?

"All models are wrong, some are useful"

1 **Probability Models** are used in the place of the real processes which gen-
2 erate the data.

3 A key feature distinguishing these from **deterministic models** is that they
4 allow for studying the variability in the system, and from this we can cal-
5 culate expected values and the probabilities of events of interest:

6
$$P(A) = \text{the probability of event } A$$

7 A recurring challenge is to make probability models sufficiently simple to
8 mathematically tractable while being sufficiently realistic to be useful.

9 **Example:** “Jumps” can be incorporated into a model for a stock price to
10 make it more realistic than a model with continuous paths.

11 See, for example, Kou (2002).

A Jump-Diffusion Model for Option Pricing

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Brownian motion and normal distribution have been widely used in the Black-Scholes option-pricing framework to model the return of assets. However, two puzzles emerge from many empirical investigations: the leptokurtic feature that the return distribution of assets may have a higher peak and two (asymmetric) heavier tails than those of the normal distribution, and an empirical phenomenon called “volatility smile” in option markets. To incorporate both of them and to strike a balance between reality and tractability, this paper proposes, for the purpose of option pricing, a double exponential jump-diffusion model. In particular, the model is simple enough to produce analytical solutions for a variety of option-pricing problems, including call and put options, interest rate derivatives, and path-dependent options. Equilibrium analysis and a psychological interpretation of the model are also presented.

(Contingent Claims; High Peak; Heavy Tails; Interest Rate Models; Rational Expectations; Overreaction and Underreaction)

1. Introduction

Despite the success of the Black-Scholes model based on Brownian motion and normal distribution, two empirical phenomena have received much attention recently: (1) the asymmetric leptokurtic features—in other words, the return distribution is skewed to the left, and has a higher peak and two heavier tails than those of the normal distribution, and (2) the volatility smile. More precisely, if the Black-Scholes model is correct, then the implied volatility should be constant. In reality, it is widely recognized that the implied volatility curve resembles a “smile,” meaning it is a convex curve of the strike price.

Many studies have been conducted to modify the Black-Scholes model to explain the two empirical phenomena. To incorporate the asymmetric leptokurtic features in asset pricing, a variety of models have been proposed:¹ (a) chaos theory,

fractal Brownian motion, and stable processes; see, for example, Mandelbrot (1963), Rogers (1997), Samorodnitsky and Taqqu (1994); (b) generalized hyperbolic models, including $\log t$ model and log hyperbolic model; see, for example, Barndorff-Nielsen and Shephard (2001), Blattberg and Gonedes (1974); (c) time-changed Brownian motions; see, for example, Clark (1973), Madan and Seneta (1990), Madan et al. (1998), and Heyde (2000). An immediate problem with these models is that it may be difficult to obtain analytical solutions for option prices. More precisely, they might give some analytical formulae for standard European call and put options, but any analytical solutions for interest rate derivatives and path-dependent options, such as perpetual American options, barrier, and lookback options, are unlikely.

In a parallel development, different models are also proposed to incorporate the “volatility smile” in option pricing. Popular ones include: (a) stochastic volatility and ARCH models; see, for example, Hull and White (1987), Engle (1995), Fouque et al. (2000); (b) constant elasticity model (CEV) model; see,

¹ Although most of the studies focus on the leptokurtic features under the physical measure, it is worth mentioning that the leptokurtic features under the risk-neutral measure(s) lead to the “volatility smiles” in option prices.

for example, Cox and Ross (1976), and Davydov and Linetsky (2001); (c) normal jump models proposed by Merton (1976); (d) affine stochastic-volatility and affine jump-diffusion models; see, for example, Heston (1993), and Duffie et al. (2000); (e) models based on Lévy processes; see, for example, Geman et al. (2001) and references therein; (f) a numerical procedure called “implied binomial trees”; see, for example, Derman and Kani (1994) and Dupire (1994). Aside from the problem that it might not be easy to find analytical solutions for option pricing, especially for path-dependent options (such as perpetual American options, barrier, and lookback options), some of these models may not produce the asymmetric leptokurtic feature (see §2.3).

The current paper proposes a new model with the following properties: (a) It offers an explanation for two empirical phenomena—the asymmetric leptokurtic feature, and the volatility smile (see §§3 and 5.3). (b) It leads to analytical solutions to many option-pricing problems, including European call and put options (see §5); interest rate derivatives, such as swaptions, caps, floors, and bond options (see §5.3 and Glasserman and Kou 1999); path-dependent options, such as perpetual American options, barrier, and lookback options (see §2.3 and Kou and Wang 2000, 2001). (c) It can be embedded into a rational expectations equilibrium framework (see §4). (d) It has a psychological interpretation (see §2.2).

The model is very simple. The logarithm of the asset price is assumed to follow a Brownian motion plus a compound Poisson process with jump sizes double exponentially distributed. Because of its simplicity, the parameters in the model can be easily interpreted, and the analytical solutions for option pricing can be obtained. The explicit calculation is made possible partly because of the memoryless property of the double exponential distribution.

The paper is organized as follows. In §2, the model is proposed, is evaluated by four criteria, and is compared with other alternative models. Section 3 studies the leptokurtic feature. A rational expectations equilibrium justification of the model is given in §4. Some preliminary results, including the Hh functions, are given in §5.1. Formulae for option-pricing problems, including options on futures, are provided in §5.2.

The “volatility smiles” phenomenon is illustrated in §5.3. The final section discusses some limitations of the model.

2. The Model

2.1. The Model Formulation

The following dynamic is proposed to model the asset price, $S(t)$, under the physical probability measure $\mathbb{P}$:

$$\frac{dS(t)}{S(t-)} = \mu dt + \sigma dW(t) + d\left(\sum_{i=1}^{N(t)} (V_i - 1)\right), \quad (1)$$

where $W(t)$ is a standard Brownian motion, $N(t)$ is a Poisson process with rate λ , and $\{V_i\}$ is a sequence of independent identically distributed (i.i.d.) nonnegative random variables such that $Y = \log(V)$ has an asymmetric double exponential distribution^{2,3} with the density

$$f_Y(y) = p \cdot \eta_1 e^{-\eta_1 y} 1_{[y \geq 0]} + q \cdot \eta_2 e^{\eta_2 y} 1_{[y < 0]}, \\ \eta_1 > 1, \eta_2 > 0,$$

where $p, q \geq 0, p + q = 1$, represent the probabilities of upward and downward jumps. In other words,

$$\log(V) = Y \stackrel{d}{=} \begin{cases} \xi^+, & \text{with probability } p \\ -\xi^-, & \text{with probability } q \end{cases}, \quad (2)$$

where ξ^+ and ξ^- are exponential random variables with means $1/\eta_1$ and $1/\eta_2$, respectively, and the notation $\stackrel{d}{=}$ means equal in distribution. In the model, all sources of randomness, $N(t)$, $W(t)$, and Y s, are assumed to be independent, although this can be relaxed, as will be suggested in §2.2. For notational simplicity and in order to get analytical solutions for various option-pricing problems, the drift μ and the volatility σ are assumed to be constants, and the Brownian motion and jumps are assumed to be

² If $\eta_1 = \eta_2$ and $p = 1/2$, then the double exponential distribution is also called “the first law of Laplace” (proposed by Laplace in 1774), while the “second law of Laplace” is the normal density.

³ Ramezani and Zeng (1999) independently propose the same jump-diffusion model from an econometric viewpoint as a way of improving the empirical fit of Merton’s normal jump-diffusion model to stock price data.

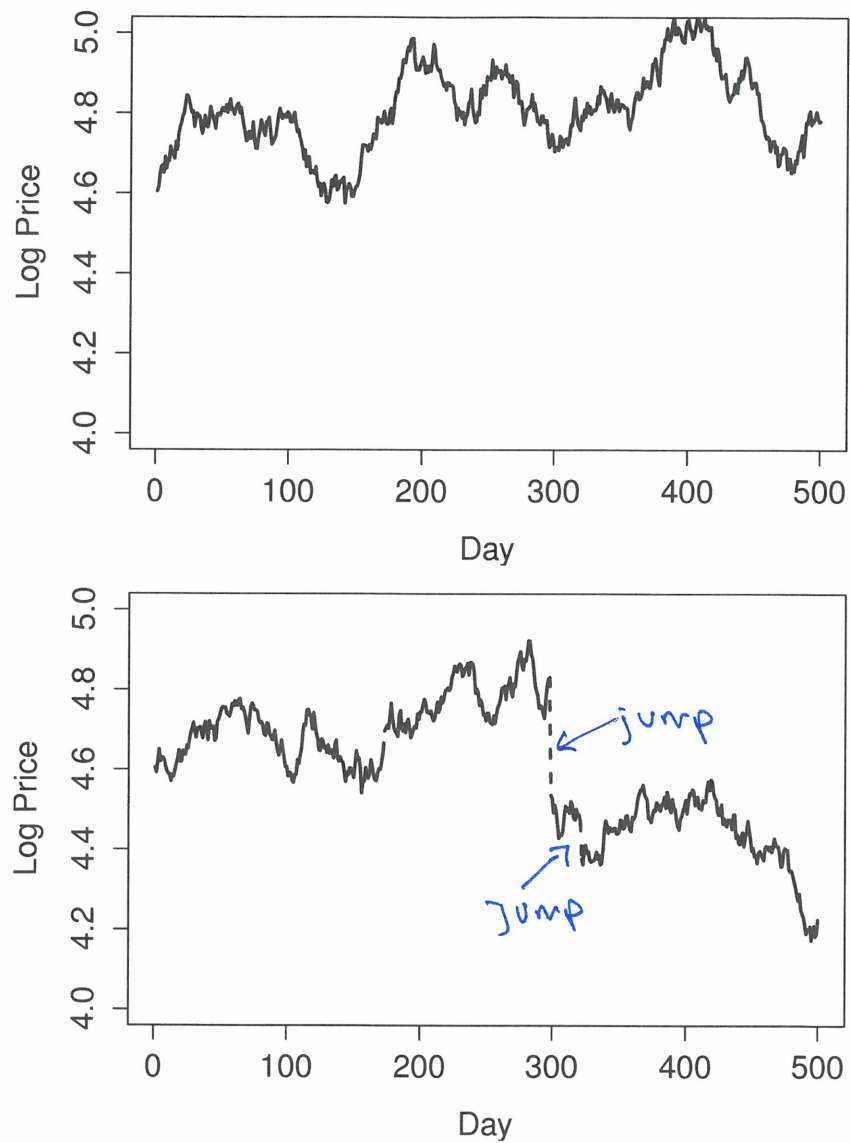
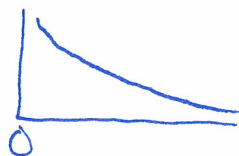


Figure 2: Examples of Brownian motion paths from a processes with and without jumps.

Comments on Kou:

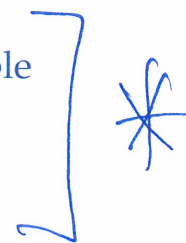
1. Note how the author balances realism and mathematical practicality with this choice of model.
2. **Model validation** is a crucial step, consisting of comparing your model with real data.
3. Note the importance of understanding a wide range of probability distributions.



Exercise: One (of multiple) components to this model is the distribution that specifies the jump sizes. One puts forth a model in the form of

The size of a jump (denote it X) is modelled as a random variable with density $f(\cdot)$, i.e.,

$$P(a \leq X \leq b) = \int_a^b f(u) du, \quad a \leq b.$$



What assumption does Kou make regarding $f(\cdot)$? Does this assumption, by itself, enable the calculation of all probabilities regarding jumps?

Assumes $f(\cdot)$ is double exponential

$$f(x) = \frac{\lambda}{2} e^{-\lambda|x|}, \quad -\infty < x < \infty$$

As stated, the model* only specifies the prob. for a limited collection of events. Denote this collection $\mathcal{F}$. I.e., if $A \in \mathcal{F}$, then $P(A)$ is defined.

For example,

$A = \{\text{first jump is positive}\}$

$B = \{\text{second jump greater than } 0.2\}$

then $P(A), P(B)$ defined, so $A \in \mathcal{F}, B \in \mathcal{F}$

But, $C = \{\text{the first two jumps are positive}\}$

$C \notin \mathcal{F}$. If I want to claim

$P(C) = P(A)^2$, then assuming that

jumps are independent. As * is written, this is not given.

The point: Enlarging $\mathcal{F}$ requires additional assumptions. Are these realistic?

Formally, a probability model consists of three parts:

1. The **sample space** (Ω), the collection of all possible **outcomes** for the probability experiment. $\omega \in \Omega$

2. A collection of subsets of Ω , called the **events** and denoted $\mathcal{F}$, for which the probability is defined.

3. The **probability measure** (P) assigns probabilities to elements of $\mathcal{F}$.

$$P(A)$$

Often write $(\Omega, \mathcal{F}, P)$ as the **probability space**.

Exercise: Does $\mathcal{F}$ contain every subset of Ω ?

In general, no. See example above.

We say $\mathcal{F} \subseteq 2^\Omega \leftarrow$ "power set"
all subsets of Ω

E.g. $\mathcal{F}_t$ = collection of events at time t

$\mathcal{F}_t \subseteq \mathcal{F}_{t+1} \subseteq \mathcal{F}_{t+2} \subseteq \dots \leftarrow$ "filtration"

Assigning Probabilities to Events

In some cases, $P(A)$ is “known” or “given” or “obvious.” (But, be careful with your intuition.)

In other cases, $P(A)$ is found using “rules.”

However it is done, the probability measure must satisfy the following:

1. $P(A) \geq 0$ for all $A \in \mathcal{F}$

2. $P(\Omega) = 1$

3. If $A_1, A_2, \dots$, are disjoint events ($A_i \cap A_j = \emptyset$ for $i \neq j$), then

$$P\left(\bigcup_{i=1}^{\infty} A_i\right) = \sum_{i=1}^{\infty} P(A_i).$$

Set-Theoretic Probability Rules

1. **Equally-Likely Outcomes:** If Ω is a finite collection of outcomes and these outcomes are **equally-likely**, then $\mathcal{F} = 2^\Omega$ and for any $A \in \mathcal{F}$,

$$P(A) = \frac{\text{number of elements in } A}{\text{number of elements in } \Omega}.$$

2. **The Addition Rule:** For any events A, B ,

$$P(A \text{ or } B) = P(A \cup B) = P(A) + P(B) - P(A \cap B).$$

3. **The Rule for Complements:** For any event A ,

$$P(A^c) = 1 - P(A).$$

Exercise: Suppose I toss a (fair, six-sided) die twice. What is the probability that the sum of the two tosses is seven?

Point: Outcomes are not necessarily equally likely. If $\Omega = \{2, 3, \dots, 11\}$ and I assumed incorrectly ELO, then conclude $P(\text{sum is } 7) = 1/11$, but this is incorrect.

Instead,

$$\Omega = \left\{ \begin{array}{l} (1,1), (1,2), \dots, (1,6), \\ (2,1), \dots, (2,6), \\ \vdots \\ (6,1), \dots, (6,6) \end{array} \right\}$$

Have 36 ELO, 6 of them yield a sum of 7, so $P(\text{sum is } 7) = 6/36$

Exercise (From *Heard on the Street*, 12th Edition, by Crack): You are dealt exactly two playing cards from a well-shuffled standard 52-card deck. The standard deck contains exactly four Kings. What is the probability that both of your cards are Kings?

Use ELO.

$$\binom{n}{k} = n C_k = \text{"n choose k"}$$

= # of groups of size k that
can be formed from n distinct elements

$$= \frac{n!}{k!(n-k)!}$$

There are $\binom{52}{2}$ possible two-card hands.
They are equally likely.

How many consist of two kings?

$$(k_1, k_2), (k_1, k_3), \dots, (k_3, k_4)$$

or $\binom{4}{2} = 6$

So, the prob. is $\frac{\binom{4}{2}}{\binom{52}{2}}$ using ELO

Exercise (From HOTS): Four cards are shuffled and placed face down in front of you. Their faces (hidden) display the four elements: water, earth, wind and fire. You are to turn the cards over, one at a time, until you either win or lose. You win if you turn over water and earth. You lose if you turn over fire. What is the probability that you win?

Recognize that "wind" is irrelevant

There are $3! = 6$ possible orderings, they are ELO

Two of them have water and earth before fire
 (H_2O, e, f) , (e, H_2O, f)

So, prob. of winning is $\frac{2}{6}$.

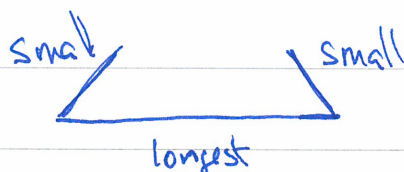
Or: win iff fire is in 3rd position.
Equally likely to be in each position,
so prob. is $\frac{1}{3}$

Exercise (From HOTS): Take a stick and break it randomly into three pieces (i.e., two randomly placed breaks on the stick). What is the probability you can form a triangle from the pieces?



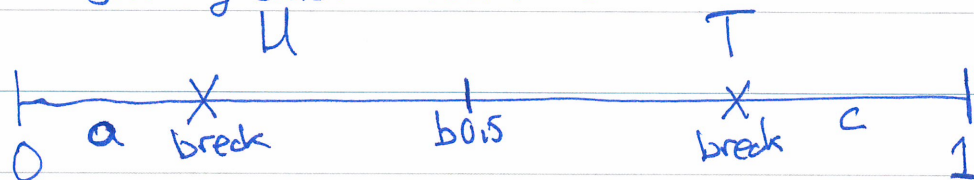
Assume positions are uniform between 0 and 1

Can form a Δ iff sum of lengths of two smaller pieces is greater than length of longest



Cannot form a Δ iff longest side has length greater than 0.5

i.e. Cannot form a Δ iff any piece has length greater than 0.5

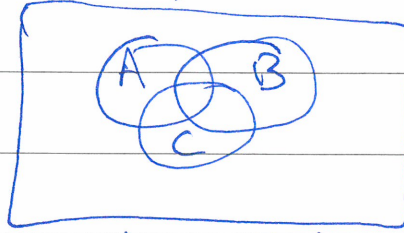


$A = \{ \text{segment } a \text{ has length at least } 0.5 \}$

$B = \{ \text{" } b \text{ " " " " " " } \}$

$C = \{ \text{" } c \text{ " " " " " " } \}$

$$\begin{aligned}
 P(\text{cannot form a } \Delta) &= P(A \cup B \cup C) \\
 &= P(A) + P(B) + P(C) - P(A \cap B) - P(A \cap C) \\
 &\quad - P(B \cap C) + P(A \cap B \cap C)
 \end{aligned}$$

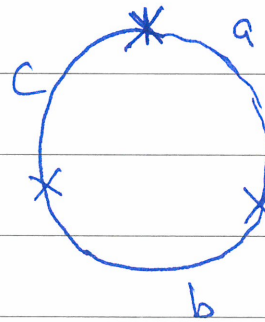


$$= P(A) + P(B) + P(C) \quad \text{since intersections have prob. 0}$$

$$\begin{aligned}
 \text{Now, } P(A) &= P(\text{segment } a \text{ has length at least } 0.5) \\
 &= P(\text{both breaks are greater than } 0.5) \\
 &= 1/4 \quad (\text{they are like coin flips})
 \end{aligned}$$

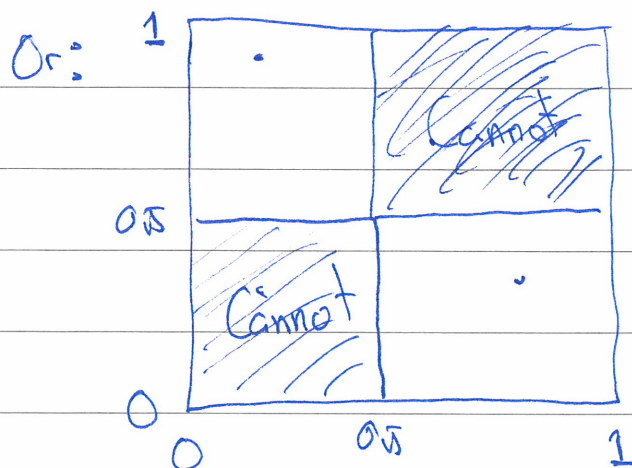
$$P(C) = 1/4$$

$$P(B) = 1/4$$



$$P(\text{Cannot form a } \Delta) = 3/4$$

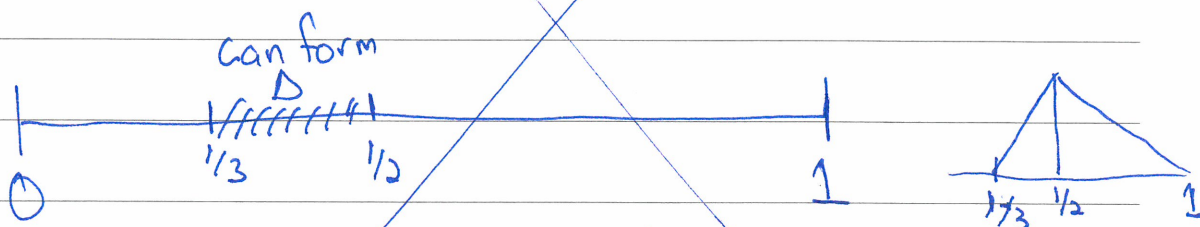
$$P(\text{Can form a } \Delta) = 1/4$$



A solution?

The longest side has length between $\frac{1}{3}$ and 1.

Can form a Δ iff longest side has length less than $\frac{1}{2}$



$$P(\text{can form a } \Delta) = \frac{\frac{1}{2} - \frac{1}{3}}{1 - \frac{1}{3}} = \frac{1}{4}$$

Assuming that length of longest side is Uniform on $\frac{1}{3}$ to 1. Not true

Conditional Probability Rules

1. **Rule for Conditional Probability:** For any events A, B with $P(B) > 0$,

$$P(A|B) = \frac{P(A \cap B)}{P(B)}.$$

"A given B"

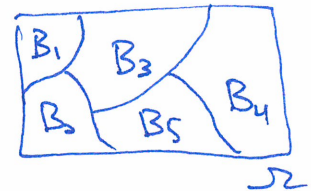
2. **Multiplication Rule:** For any events A, B ,

$$P(A \text{ and } B) = P(A \cap B) = P(A|B)P(B).$$

$$\neq P(A)P(B)$$

3. **Law for Total Probability:** Suppose events $B_1, B_2, \dots, B_n$ form a **partition** of Ω . Then for any event A ,

$$P(A) = \sum_{i=1}^n P(A|B_i)P(B_i).$$



4. **Bayes' Theorem:** Suppose events $B_1, B_2, \dots, B_n$ form a partition of Ω .

Then for any event A ,

$$P(B_i|A) = \frac{P(A|B_i)P(B_i)}{\sum_{j=1}^n P(A|B_j)P(B_j)}.$$

Exercise (Partially from HOTS): You are dealt exactly two playing cards from a well-shuffled standard 52-card deck. The standard deck contains exactly four Kings. (a) What is the probability that both of your cards are Kings? (b) What is the probability that the second card you are dealt is a King?

$$(a) \quad A = \{ \text{first card is K} \}$$
$$B = \{ \text{second " " " } \}$$

$$\begin{aligned} P(\text{both kings}) &= P(A \cap B) \\ &= P(A)P(B|A) \quad \text{using multiplication rule} \\ &= \left(\frac{4}{52}\right)\left(\frac{3}{51}\right) \end{aligned}$$

$$(b) \quad P(B) = \frac{4}{52}$$

$$P(B) = P(B|A)P(A) + P(B|A^c)P(A^c)$$

using LFTP

since A, A^c is a partition of Ω

$$\begin{aligned} &= \left(\frac{3}{51}\right)\left(\frac{4}{52}\right) + \left(\frac{4}{51}\right)\left(\frac{48}{52}\right) \\ &= \frac{4}{52} \end{aligned}$$

- 1 **Exercise (From HOTS):** You have a large jar containing 999 fair pennies
2 and one two-headed penny. Suppose you pick one coin out of the jar and
3 flip it ten times and get all heads. What is the probability that the coin you
4 chose is the two-headed one?

5 $B = \{ \text{choose the two-headed coin} \}$

6 $A = \{ \text{all 10 flips are heads} \}$

7
8 want $P(B|A)$. Try Bayes' Rule
9 because $P(A|B)$ is easy.

10
11
$$P(B|A) = \frac{P(A|B)P(B)}{P(A|B)P(B) + P(A|B^c)P(B^c)}$$

12
13 (my partition is B, B^c)

14
15
$$= \frac{\left(\frac{1}{1000}\right) \left(\frac{1}{1000}\right)}{\left(\frac{1}{1000}\right) \left(\frac{1}{1000}\right) + \left(\frac{1}{2}\right)^{10} \left(\frac{999}{1000}\right)}$$

16
17
18
19
$$= 0.51$$